AIMS African Institute for Mathematical Sciences RWANDA

Introduction to Statistics and Probability

Exercise sheet # 1

Probability

Exercise 1 – Let us consider $\Omega = \{a, b, c\}$.

1. Find $\mathscr{F}_s$, the smallest σ–algebra on Ω.
2. Find $\mathscr{F}_\ell$, the largest σ–algebra on Ω.
3. How many elements does the largest σ–algebra contain?

Exercise 2 – Let A, B, C be 3 events such that $P(A) = P(B) = P(C) = 0.3$, $P(A \cap B) = 0.15$, $P(B \cap C) = 0.15$, $P(A \cap C) = 0.2$, $P(A \cap B \cap C) = 0.08$.

1. Give the expression of the event E : A and B occur but C does not.
2. Give the expression of the event F : exactly 2 amongst the 3 events occur.
3. Calculate $P(E)$ and $P(F)$.

Exercise 3 – Let A and B be events. Assume $P(A) > 0$ and $P(B) > 0$.

1. Prove that if A, B are mutually exclusive, then A, B are not independent.
2. Prove that if A, B are independent, then A, B are not mutually exclusive.

Exercise 4 – Let A, B, C be events. Assume $P(C) > 0$ and $P(B \cap C) > 0$.
Prove or disprove each of the following :

1. $P(A \cap B) = P(A \mid B \cap C)P(B \mid C)$.
2. $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$ if A, B are independent.

If you disprove an equation, give two or more conditions under which it is true.

Exercise 5 – Consider an experiment with the following sample space

$$\Omega = \{ abc, bac, cab, bca, acb, cba, aaa, bbb, ccc \}$$

where each outcome is equally likely.
Define events $A_k = a$ in k-th position (for $k = 1,2,3$).

1. Describe each event A_k, $k = 1,2,3$.
2. Are these events mutually independent?
3. Are they independent events?

Exercise 6 – A factory has 4 machines that produce the same item. Machine 1 and 2 each produce 20 % of the total output, while machines 3 and 4 are larger and each produce 30% of the total output. It is know that 6 % of machine 1's output is defective, while machine 2 produces 5% defective items, and machine 3 and 4 each produce 8 % defective items.

An item is chosen at random from the output of the factory.

1. What is the probability that this item is defective?
2. Given that the item selected is defective, what is the probability that it was produced by machine 2?

Exercise 7 – A bin contains 100 items, of which 10% are bad. A second bin contains 100 items, of which 90% are good. Since such a large proportion of the second bin is good, by selecting an item at random from the second bin and adding it to the first bin do we increase the probability that an item chosen from the first bin will be good?